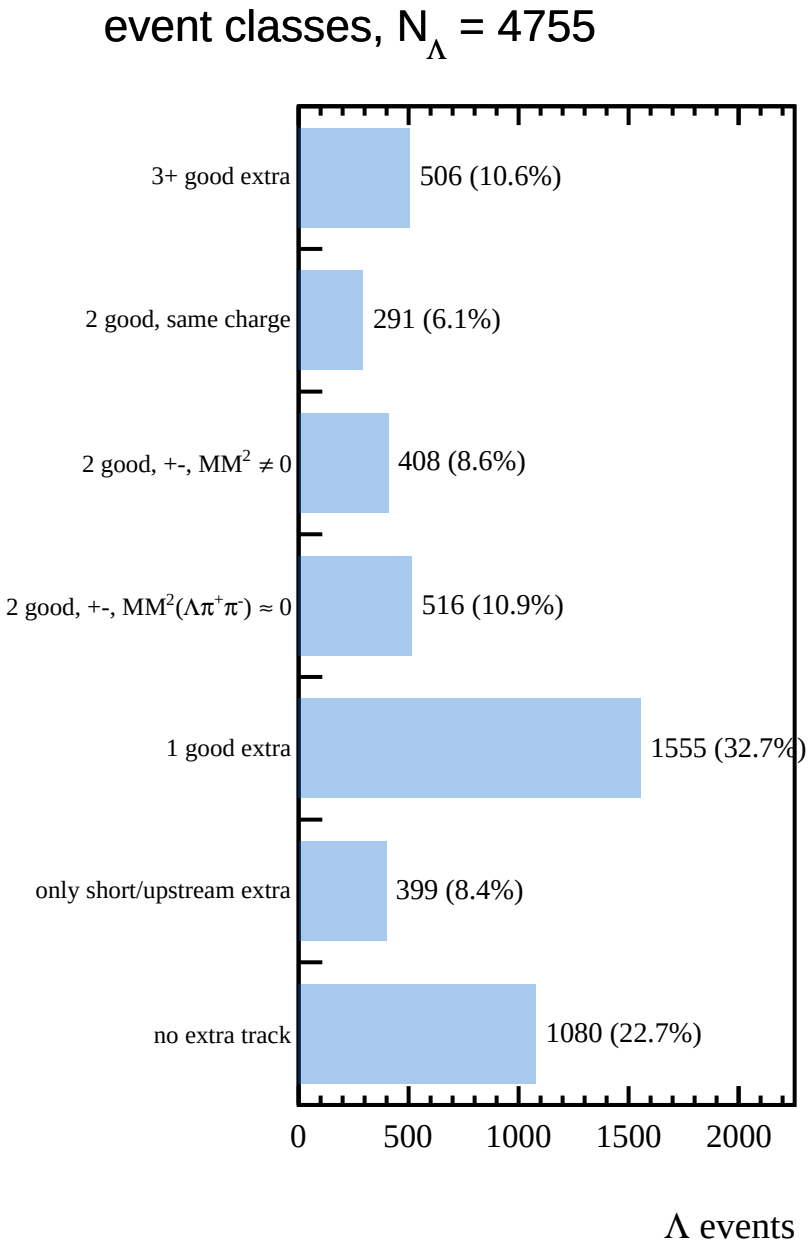
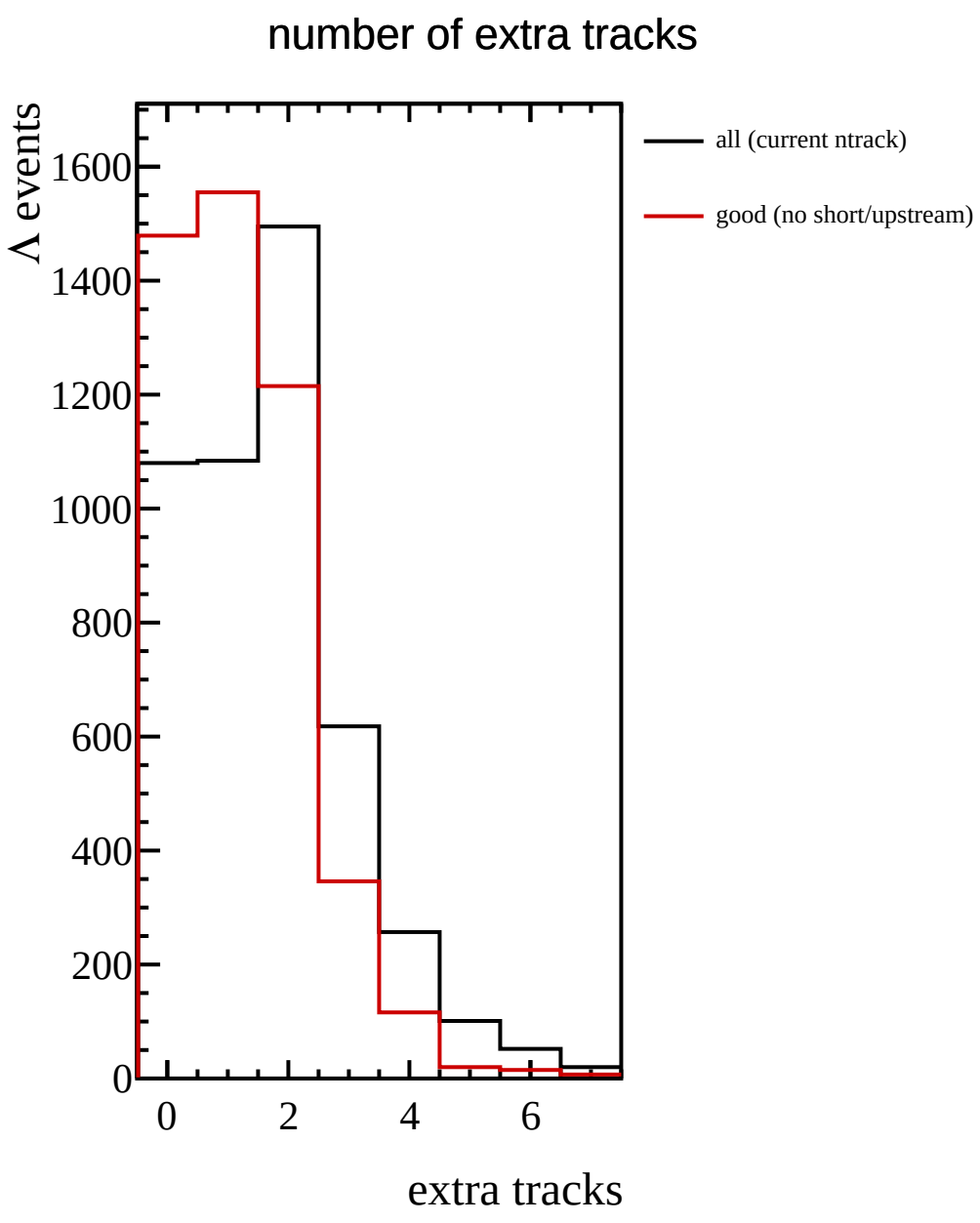
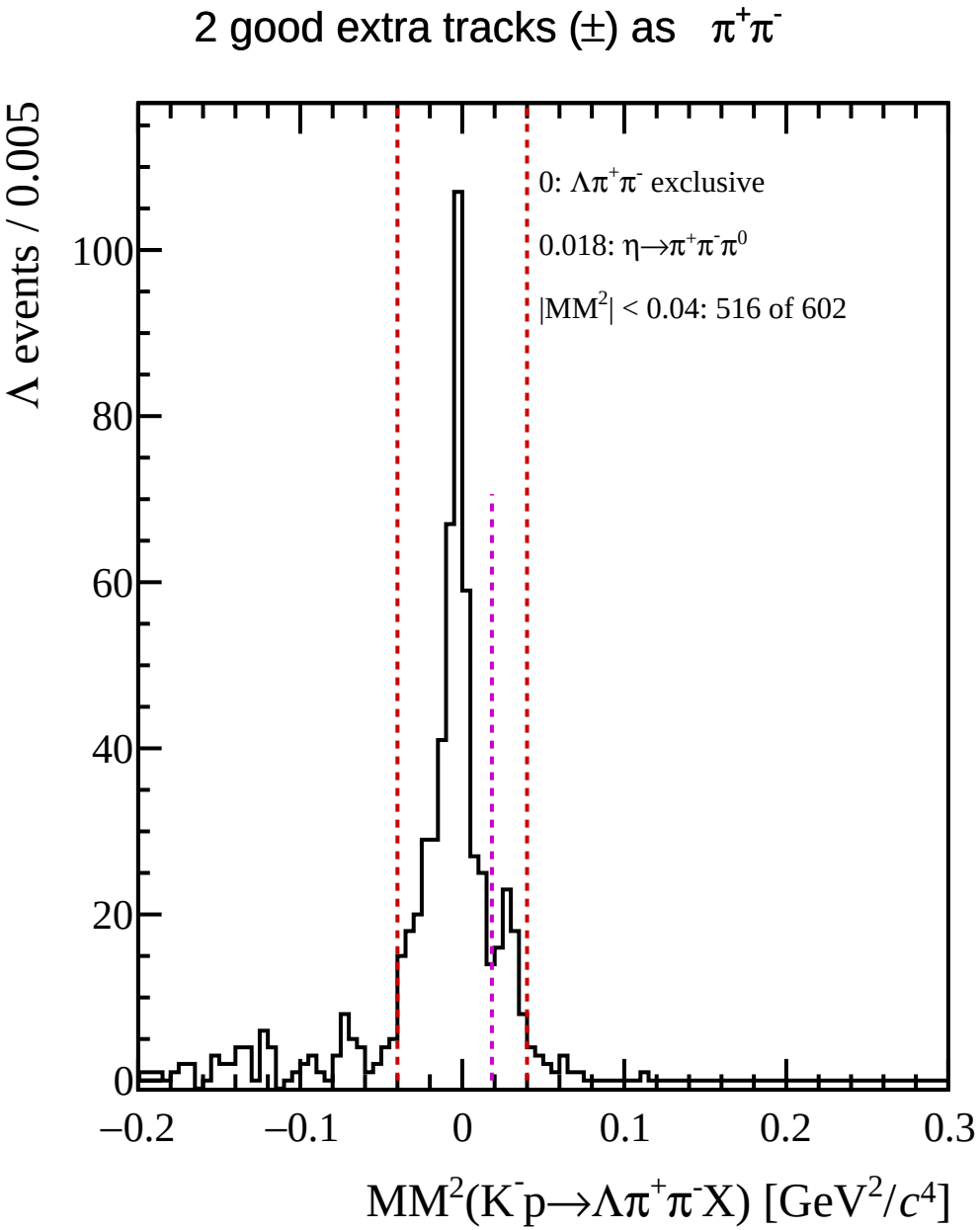
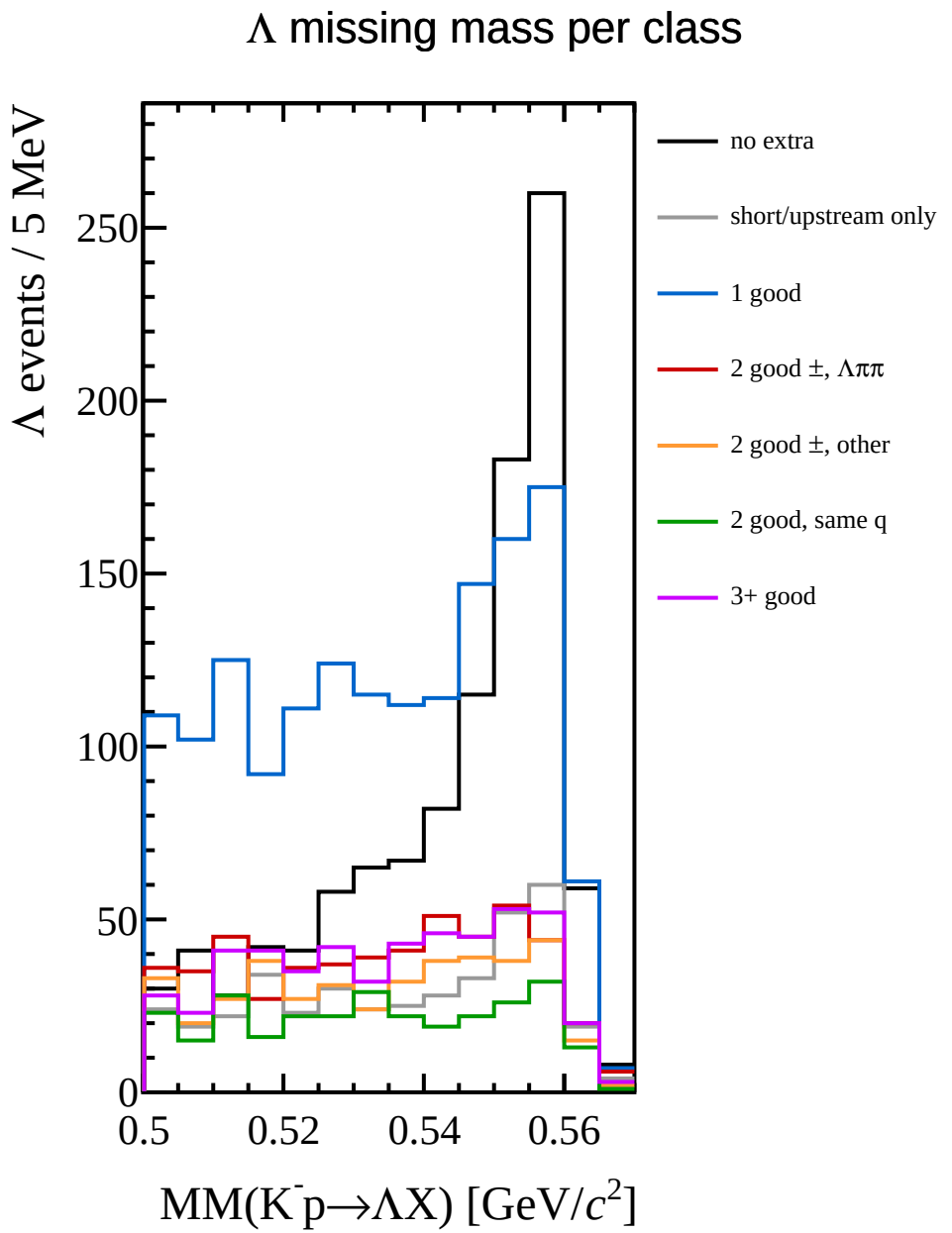
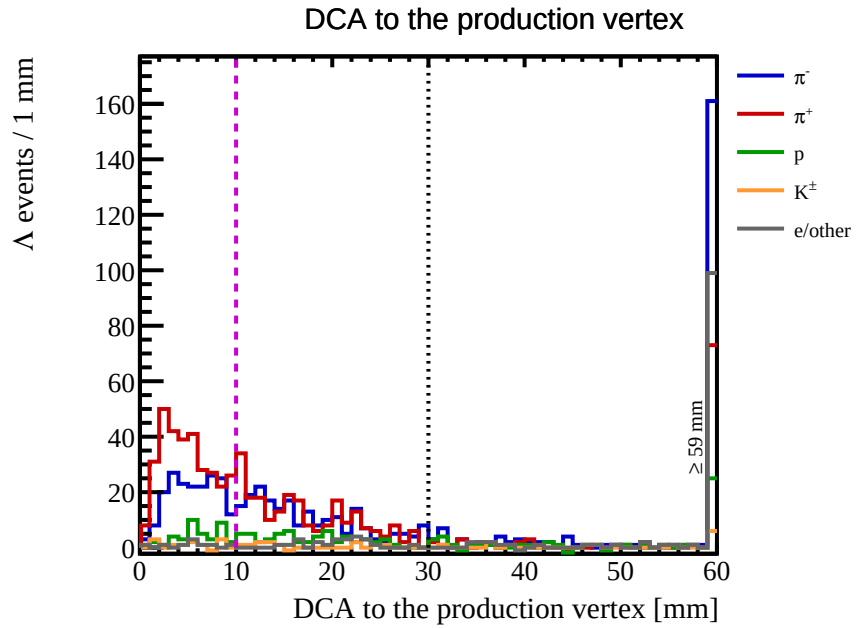

event classes, N_Λ = 4755
3+ good extra
506 (10.6%)
2 good, same charge
291 (6.1%)
2 good, +-, MM² ≠ 0
408 (8.6%)
2 good, +-, MM²(Λπ⁺π⁻) ≈ 0
516 (10.9%)
1 good extra
1555 (32.7%)
only short/upstream extra
399 (8.4%)
no extra track
1080 (22.7%)
0
500
1000
1500
2000
Λ events

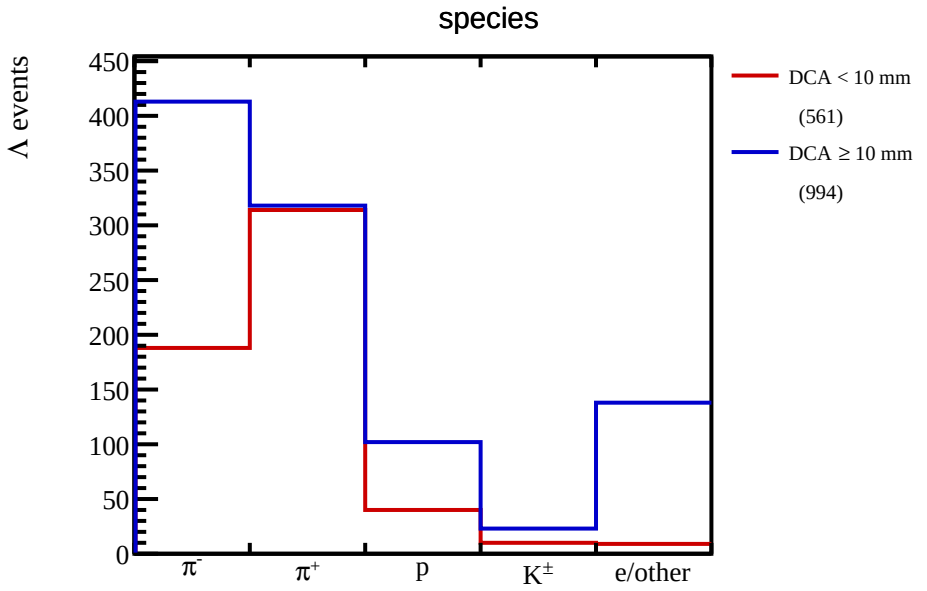

number of extra tracks
Λ events
1600
1400
1200
1000
800
600
400
200
0
0
2
4
6
extra tracks
all (current ntrack)
good (no short/upstream)

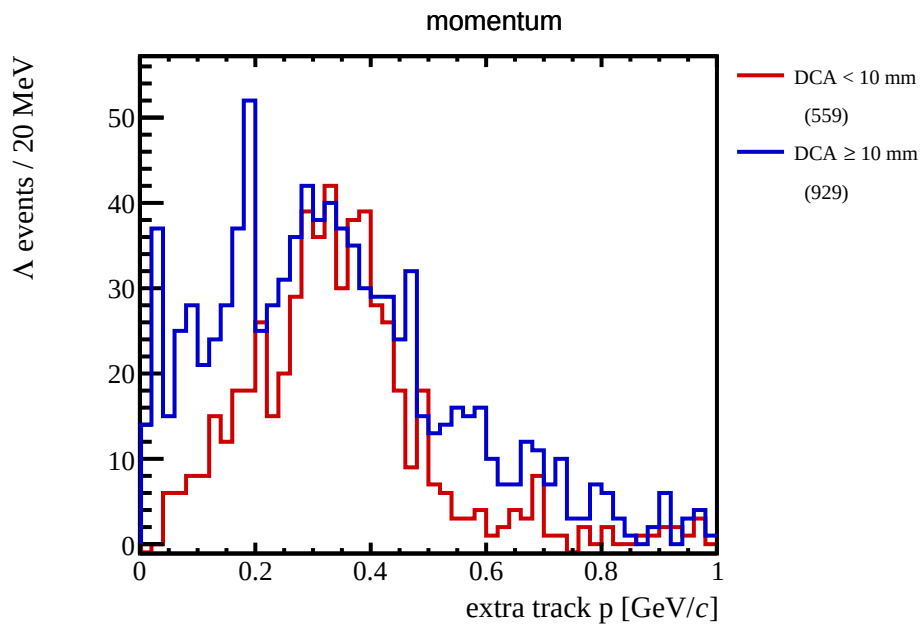

2 good extra tracks (±) as π+π-
Λ events / 0.005
0: Λπ+π- exclusive
0.018: η→π+π-π0
|MM2| < 0.04: 516 of 602
MM2(K-p→Λπ+π-X) [GeV2/c4]

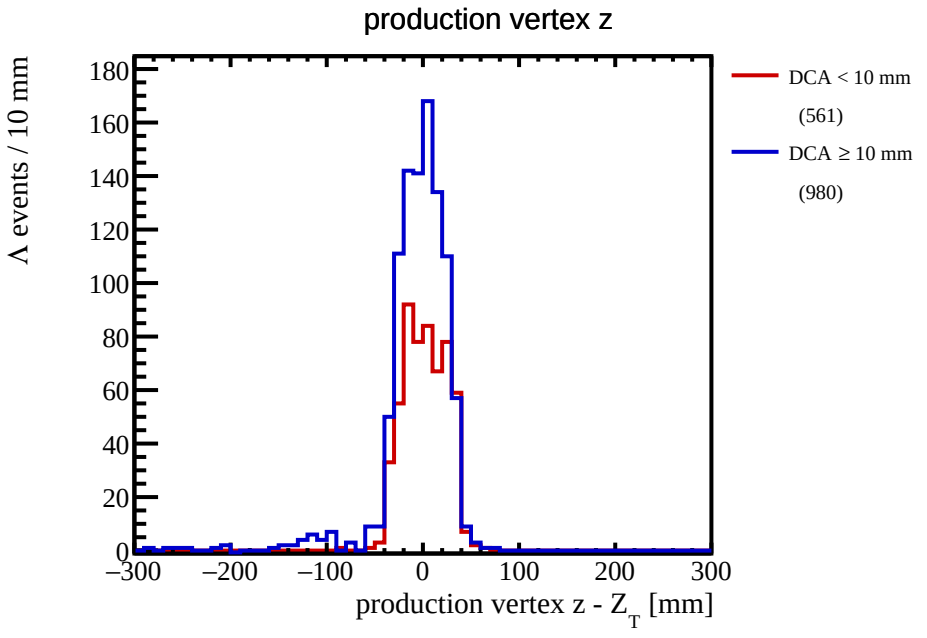

Λ missing mass per class
Λ events / 5 MeV
no extra
short/upstream only
1 good
2 good ±, Λππ
2 good ±, other
2 good, same q
3+ good
MM(K-p→ΛX) [GeV/c2]

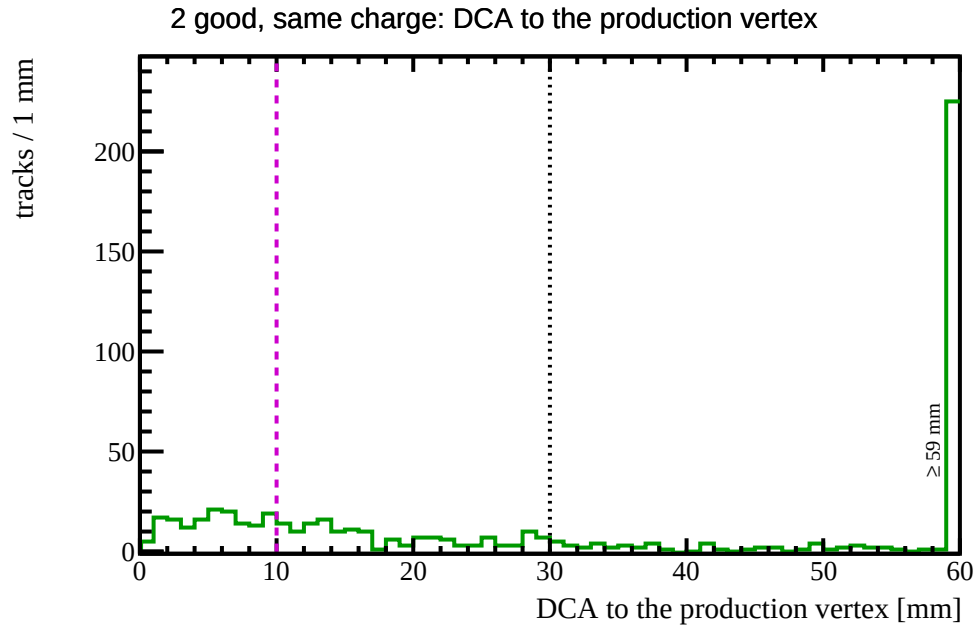
run02447-02452-n5: DCA, χ^2, LH2, MM ≥ 0.5, Λ 1.105-1.125 minus sidebands | 1 good extra track

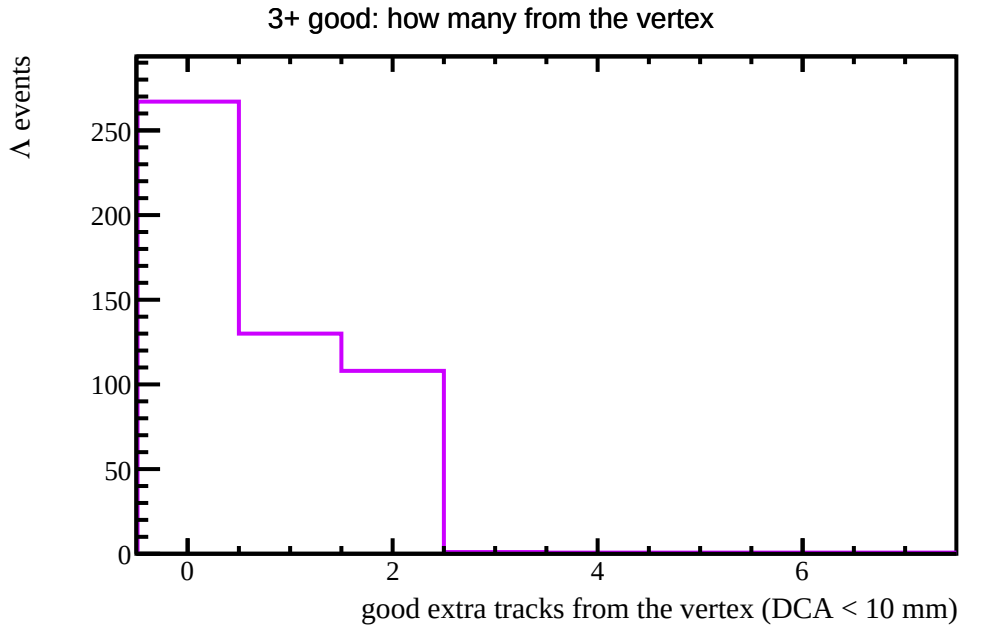


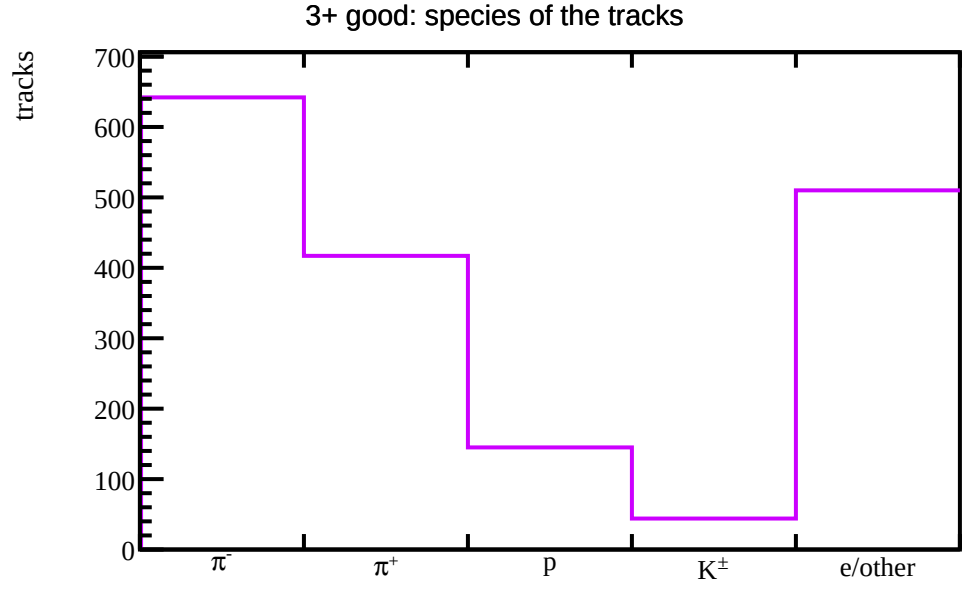


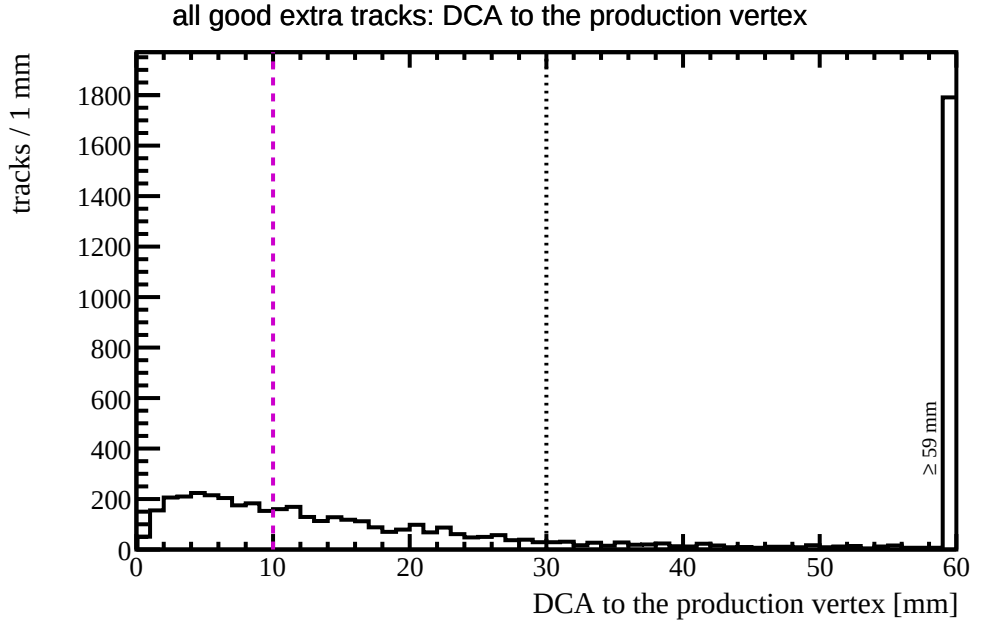


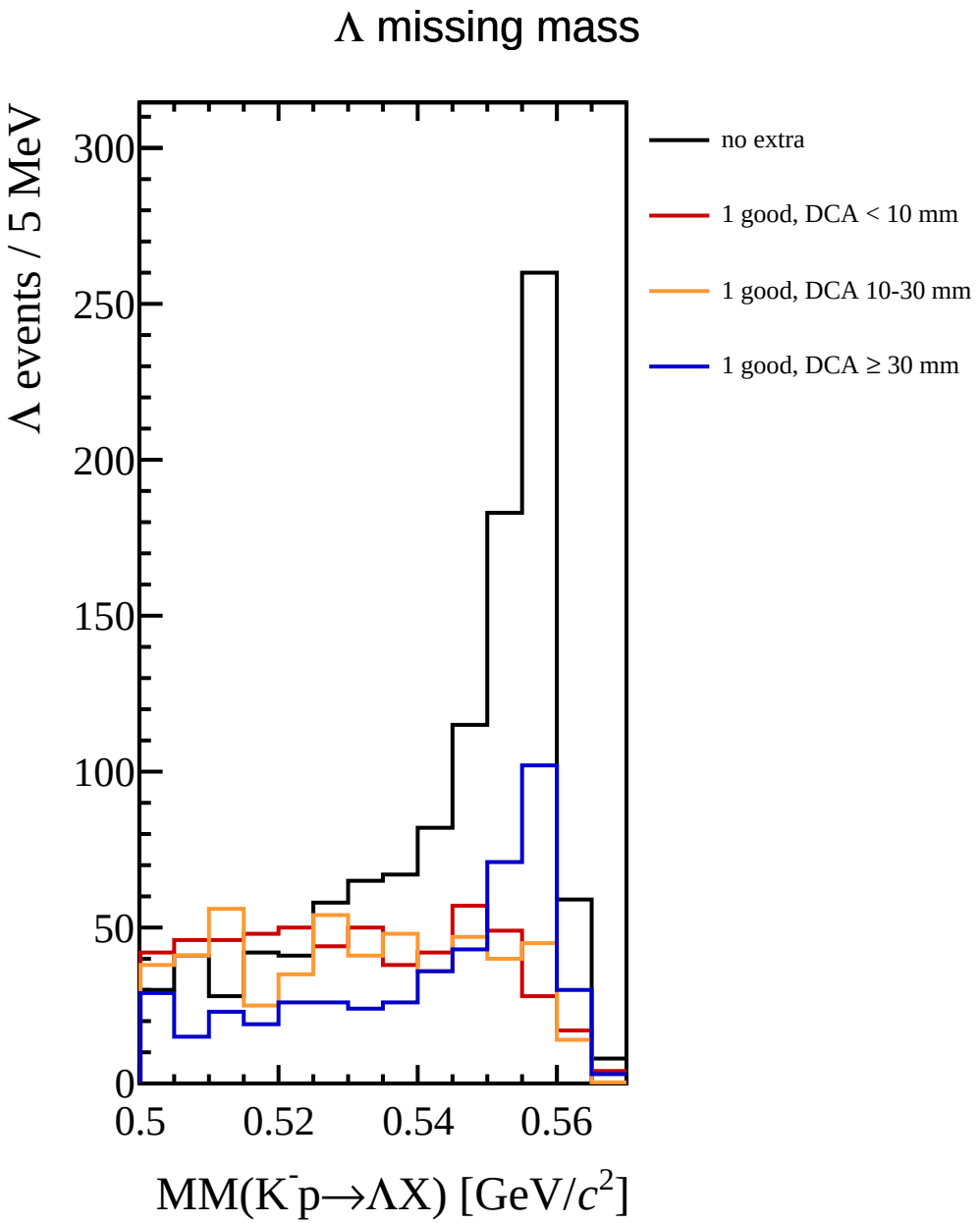


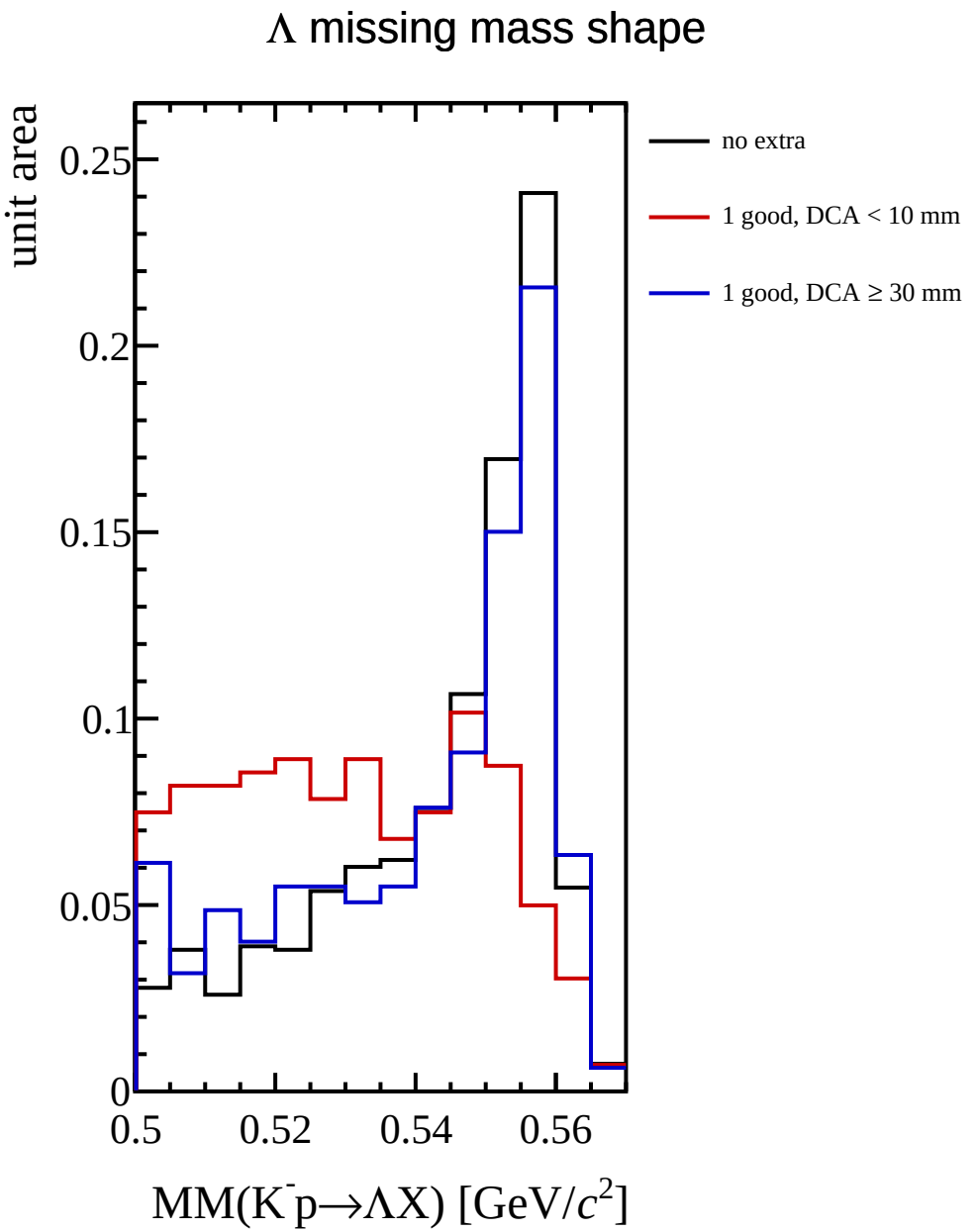

2 good, same charge: DCA to the production vertex
tracks / 1 mm
0
50
100
150
200
0
10
20
30
40
50
60
≥ 59 mm
DCA to the production vertex [mm]


3+ good: how many from the vertex
Λ events
0
50
100
150
200
250
0
2
4
6
good extra tracks from the vertex (DCA < 10 mm)

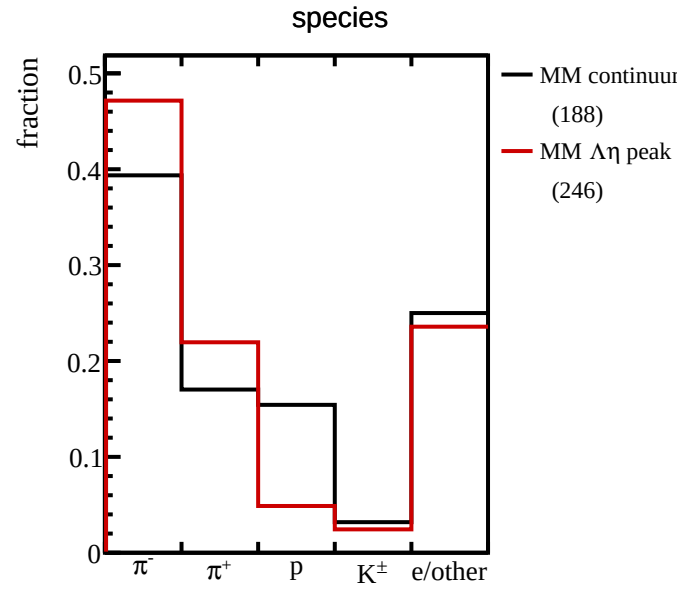

3+ good: species of the tracks
tracks
0
100
200
300
400
500
600
700
π⁻
π⁺
p
K±
e/other

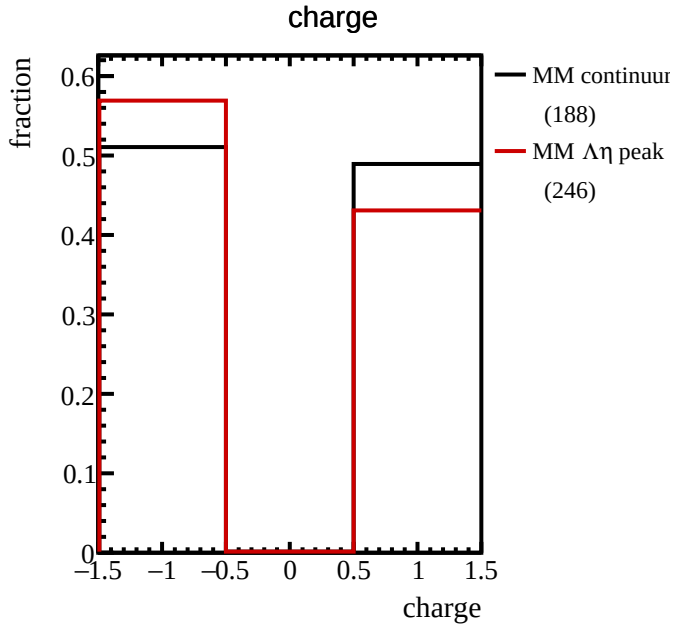

all good extra tracks: DCA to the production vertex
tracks / 1 mm
0
200
400
600
800
1000
1200
1400
1600
1800
0
10
20
30
40
50
60
≥ 59 mm
DCA to the production vertex [mm]

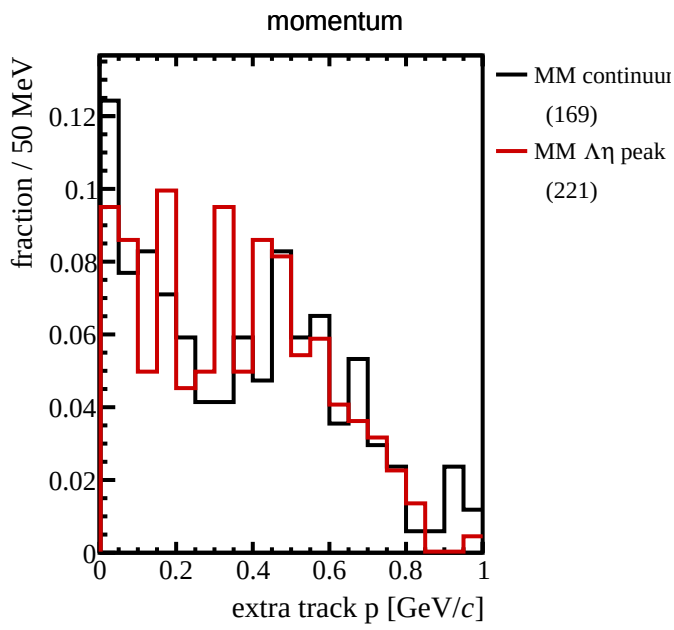

Λ missing mass
Λ events / 5 MeV
300
250
200
150
100
50
0
0.5
0.52
0.54
0.56
MM(K⁻p→ΛX) [GeV/c²]
no extra
1 good, DCA < 10 mm
1 good, DCA 10-30 mm
1 good, DCA ≥ 30 mm

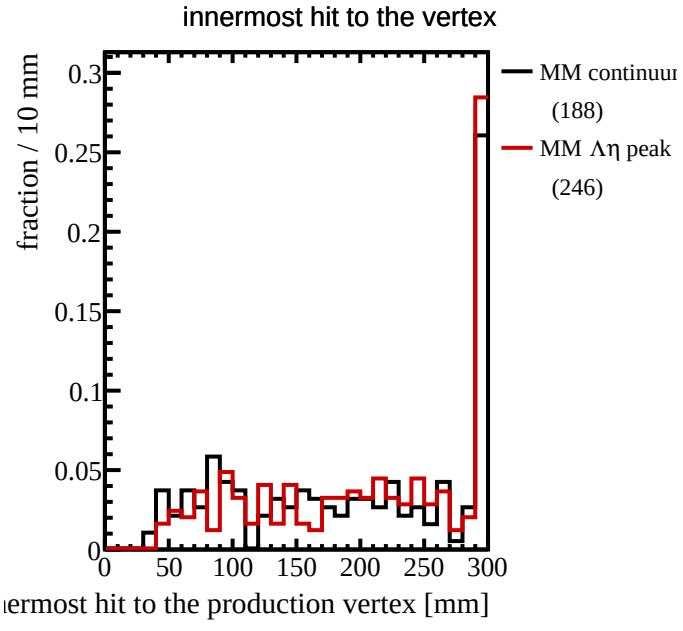

Λ missing mass shape
unit area
0.25
0.2
0.15
0.1
0.05
0
0.5
0.52
0.54
0.56
MM(K⁻p→ΛX) [GeV/c²]
no extra
1 good, DCA < 10 mm
1 good, DCA ≥ 30 mm

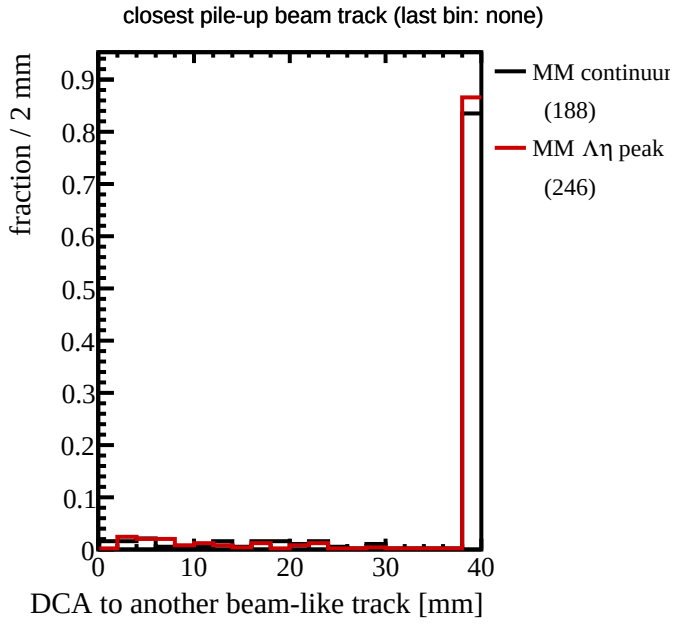
run02447-02452-n5: DCA, χ^2, LH2, MM ≥ 0.5, Λ 1.105-1.125 minus sidebands | 1 good extra track with DCA ≥ 30 mm: Λ MM 0.50-0.54 vs 0.545-0.565 (unit area)

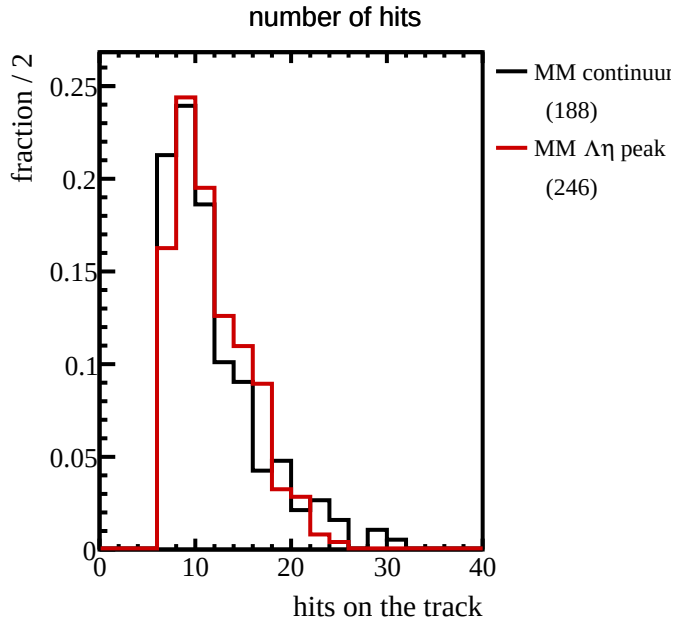


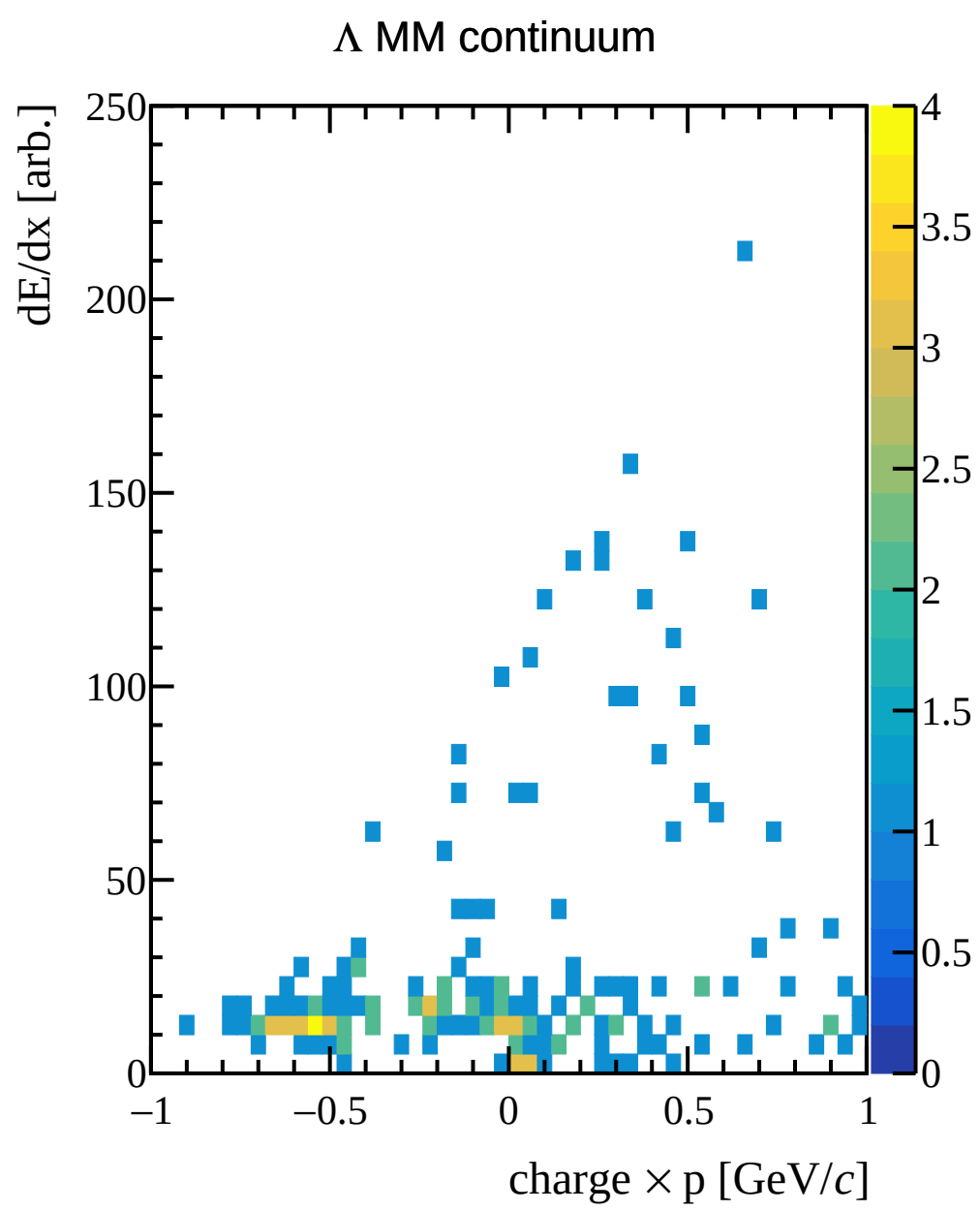


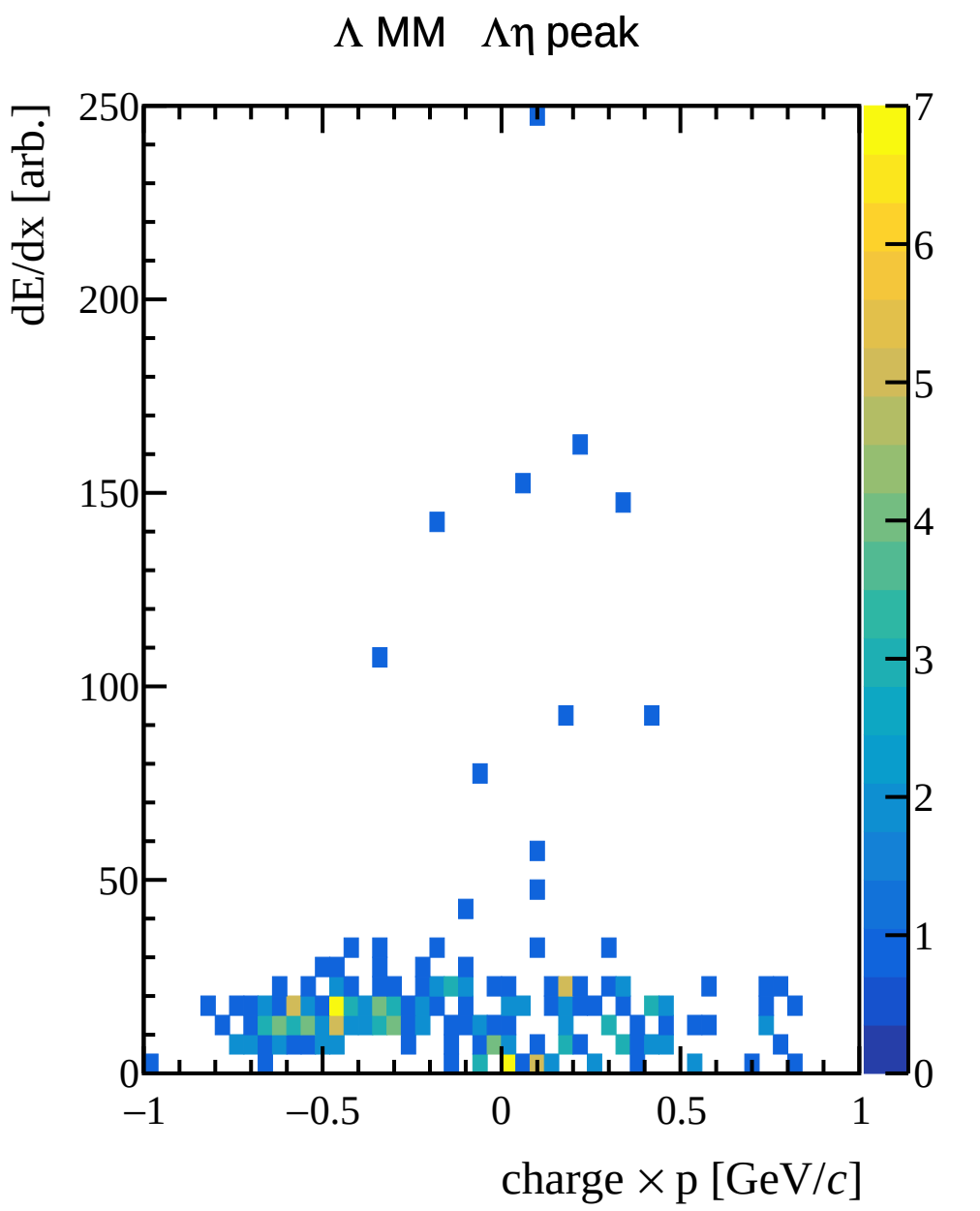









Λ MM continuum
dE/dx [arb.]
250
200
150
100
50
0
−1
−0.5
0
0.5
1
4
3.5
3
2.5
2
1.5
1
0.5
0
charge × p [GeV/c]


Λ MM Λη peak
dE/dx [arb.]
250
200
150
100
50
0
−1
−0.5
0
0.5
1
7
6
5
4
3
2
1
0
charge × p [GeV/c]